Mechanical Design Portfolio

~ Omkar Pawar

Mechanical Engineer | CAD Design | Additive Manufacturing | Automation

Portfolio showcasing mechanical design projects across **shop-floor equipment integration, metal additive manufacturing, topology optimization, robotic tooling, enclosure design, PLC automation, and simulation-driven engineering.**

Tools & Skills:
SolidWorks | ANSYS | n-Topology | Bambu Slicer | EOS M290 | FDM | LPBF | DFM | DfAM | Fixture Design | Enclosure Design | PLC/IIoT | Process Improvement | Cost Engineering | Reverse Engineering | 3D modelling | 3D Scanning | Continuous Improvement | Python | C++ | CATIA V5

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8. 3D-Printed Control Box Assembly
9. Structural and CFD Analysis of Vortex Bladeless Wind Turbine
10. Comparative Thermal Analysis of Heat Sink Fin Geometries

- ## MAARCO – Multi-terrain Amphibious ARCtic Explorer

  - **Design Type –** Robot design, drivetrain sizing, buoyancy sizing, mechanical concept development
    Tools Used: SolidWorks, engineering calculations, motor/battery sizing, material comparison, buoyancy analysis

  - **Objective**
    - Design a multi-terrain robotic platform capable of operating across **land, water, mud, and ice** using two independently driven screw legs. The goal was to size the drivetrain, power system, and flotation support for a robot carrying an estimated **~60 kg total payload**.

  - **Design Description**
    - Developed the mechanical concept for a screw-drive robot using two large helical cylindrical legs mounted on either side of the central body. Each screw leg is independently powered to support forward motion, turning, and terrain adaptability. The design was developed around a **6-inch diameter**, **4 ft long** screw leg with a **100 mm pitch**.
    - The drivetrain sizing focused on estimating the torque required at each screw leg to move the robot across land, mud, ice, and water. The robot uses two independently driven screw legs, so the total tractive effort is split between the left and right motors.
      The required drive torque was estimated from the axial driving force needed at the screw leg:
      - **$T = F \times p / (2\pi \times \eta)$**
        - Where : T = torque required at one screw leg
          F = axial/tractive force required at one screw leg
          p = screw pitch = 100 mm = 0.1 m
          η = estimated screw-terrain efficiency
    -

  - **Key Design Features**
    - Key Design Features:
      - Dual independently driven screw-leg drivetrain
      - 6 inch diameter, 4 ft long screw legs
      - 100 mm screw pitch
      - Approximate 60 kg central payload
      - Designed for land, water, mud, and ice
      - 72 V, 3000 W motor selection
      - 2 × 72 V, 30 Ah battery system
      - Buoyancy foam sizing for water operation

  - **Engineering Considerations**
    - Estimated torque demand for screw-leg motion
      - Considered slip, drag, mud resistance, and traction losses
      - Selected motors based on torque, voltage, power, and RPM requirements
      - Sized batteries based on runtime and power demand
      - Calculated the required buoyancy force using Archimedes' principle
      - Compared closed-cell polyethylene foam, cross-linked polyethylene foam, polyurethane marine flotation foam, and Blue Robotics buoyancy foam

  - **Result**
    The robot is currently in the design and prototype-development stage. Key sizing work has been completed for the screw-leg geometry, drivetrain, battery system, and flotation material selection. The project demonstrates practical mechanical design judgment across drivetrain sizing, power selection, buoyancy analysis, material comparison, and manufacturability planning.

- ## Weight-Based Batch Counting System for Aluminum Short Profiles

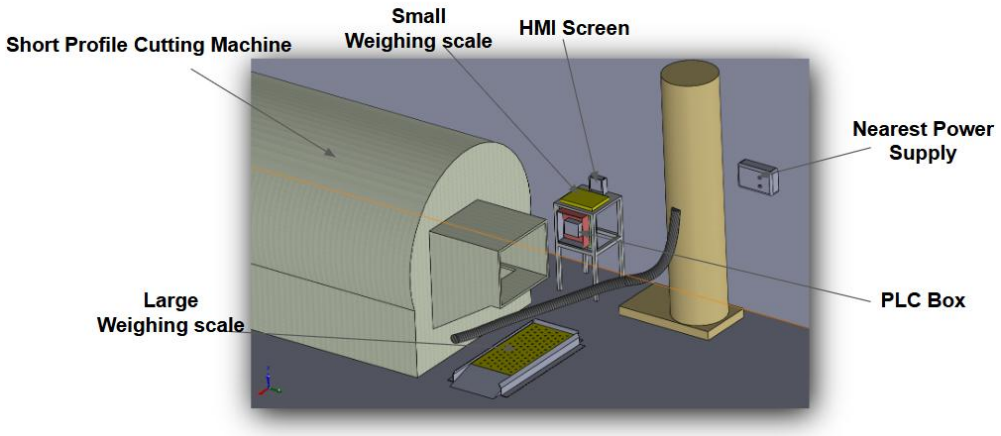
- **Design Type -** Manufacturing equipment layout, shop-floor system integration, production support hardware
  Tools Used: SolidWorks, weighing-scale system, HMI/PLC integration planning

- **Objective**
  - Design and implement a weight-based counting system to track aluminum short profiles produced at the cutting machine and reduce dependency on manual counting.

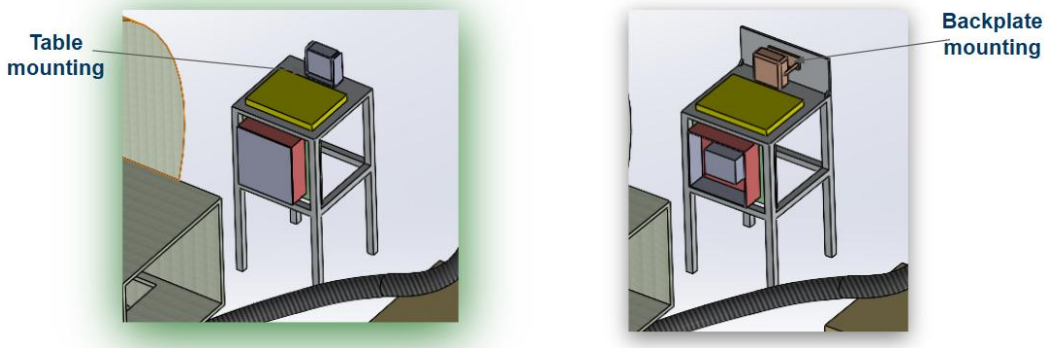
- **Design Description**
  - Developed a SolidWorks layout of the target implementation area on the shop floor, including the short-profile cutting machine, large floor scale, small weighing scale, HMI screen, PLC box, and nearest power supply. The design was created around the existing machine location, operator workflow, available floor space, and material handling path.
  - The system used a large floor scale for bulk batch weighing and a smaller precision scale for smaller quantities. Part count was calculated from net weight by accounting for tare weight and known part weight. Two HMI mounting layouts were evaluated: a table-mounted layout and a backplate-mounted layout. The backplate concept improved screen visibility, organization, and protection, while the table-mounted layout was simpler to install.

- **Key Design Features**
  - Floor-scale layout for bulk batch counting
  - Small-scale station for smaller lots to calculate the weight per piece
  - HMI location planning for operator access
  - PLC box and power supply placement in CAD layout
  - Table-mounted and backplate-mounted HMI concepts
  - Shop-floor implementation near the existing production machine
  - Future-ready concept for SAP/PLC traceability
- **Engineering Considerations**
  - Designed around the existing short-profile cutting machine location
  - Considered operator reach, workflow, scale access, and cart movement
  - Accounted for tare weight during part-count calculation
  - Evaluated scale resolution, uneven floor conditions, and vibration effects
  - Planned system architecture for future PLC/database/SAP integration
  - Used CAD layout planning before installation to reduce implementation risk

CAD Design of Targeted Implementation Area on Shop Floor



CAD Design - 2 Different Table Setups considered



Layout 1 - HMI Screen on table

Layout 2 - HMI Screen on a backplate

Weighing Scale setup in real life



- **Result**

  Implemented the weighing-scale setup on the shop floor and achieved approximately 98% counting accuracy for aluminum short-profile batch counting. The system reduced dependence on manual counting, improved batch consistency, and created a scalable foundation for future digital production tracking.

- ## Production of LPBF-based Schwarz-P Heat Sink

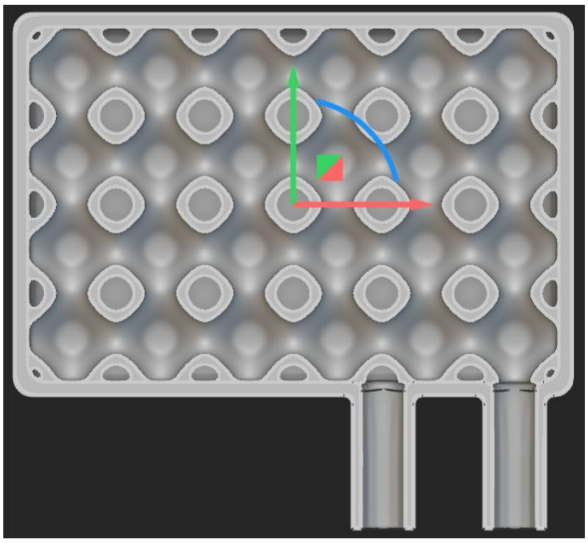
- **Design Type:** Metal additive manufacturing, thermal management, TPMS lattice design
  Tools Used: SolidWorks, nTopology, EOS M290

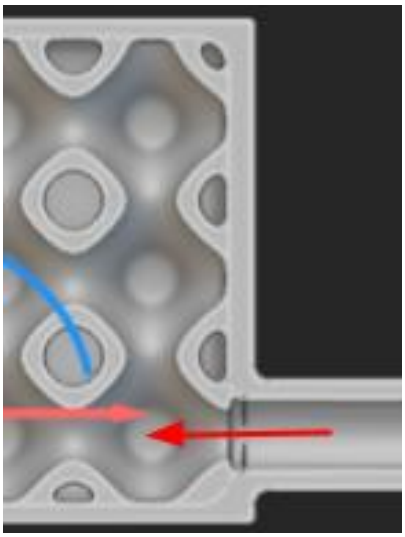
- **Objective**
  - Design, fabricate, and test a metal 3D-printed heat sink using a Schwarz-P TPMS lattice structure to improve thermal performance through increased internal surface area and fluid interaction.

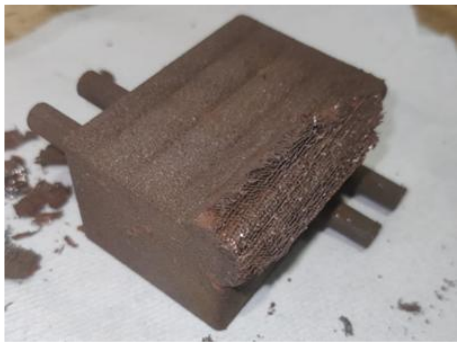
- **Design Description**
  - Designed a compact heat sink with an internal Schwarz-P triply periodic minimal surface lattice. The TPMS structure created a continuous internal surface network to increase heat transfer area while maintaining internal flow paths. The design included inlet and outlet tube features for water flow and a rectangular body for thermal contact.
  - The heat sink was modeled using SolidWorks and nTopology, then fabricated using Laser Powder Bed Fusion on an EOS M290 metal printer. GR-Cop-42 copper alloy was selected for its thermal performance and compatibility with advanced metal additive manufacturing.

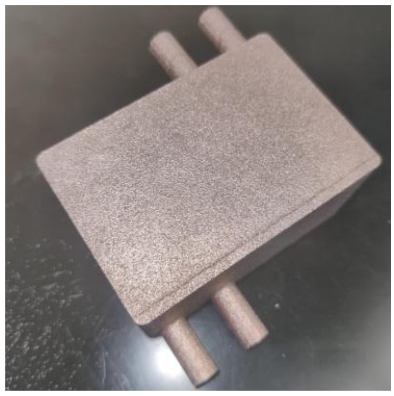
- **Manufacturing Details**
  - Material: GR-Cop-42 copper alloy
  - Process: Laser Powder Bed Fusion
  - Machine: EOS M290
  - Lattice Type: Schwarz-P TPMS
  - Laser Power: 325 W
  - Scan Speed: 825 mm/s
  - Hatch Spacing: 0.1 mm

- **Key Design Features**
  - Schwarz-P TPMS internal lattice for high surface area
  - Integrated inlet and outlet flow connections
  - Compact rectangular heat sink body
  - Internal flow paths enabled by LPBF manufacturing
  - Copper alloy material for thermal performance
  - Additive-manufacturing-driven geometry not feasible through conventional machining











- **Result**
  - Successfully fabricated and experimentally tested the heat sink. Hot-plate testing showed approximately 10.5°C coolant temperature rise over 300 seconds and approximately 12°C lower surface temperature compared with natural convection.

- ## Topology-Optimized 3D-Printed Bridge for Load-to-Weight Performance

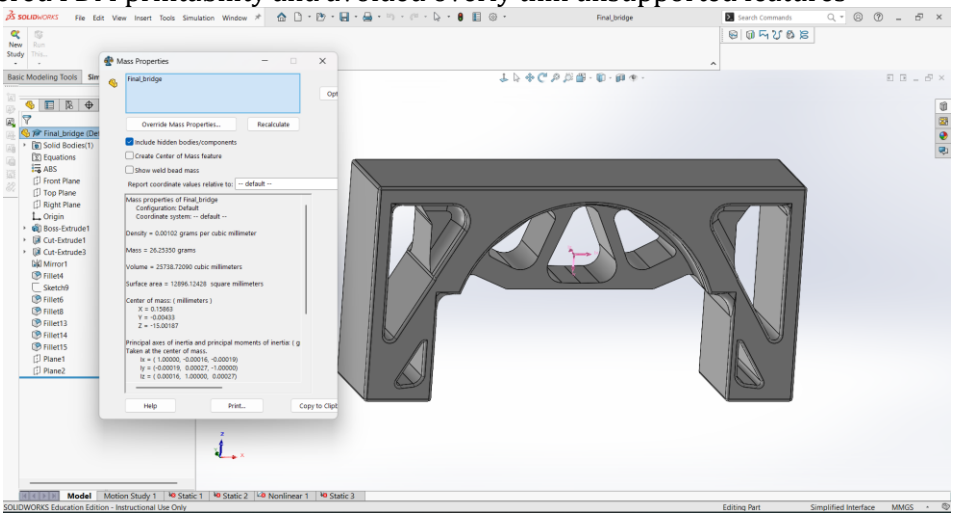
- **Design Type:** Topology optimization, structural design, FDM 3D printing
  Tools Used: SolidWorks, SolidWorks Simulation

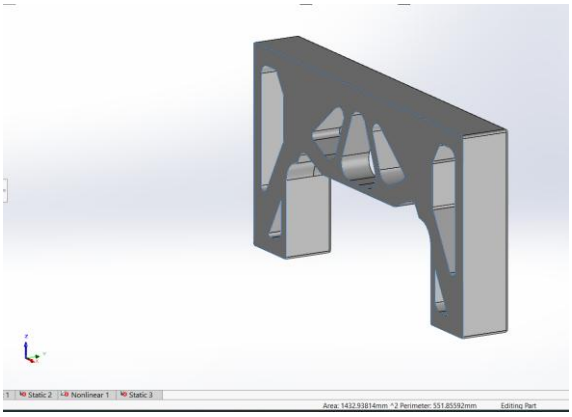
- **Objective**
  - Design a lightweight 3D-printed bridge structure that maximizes load-bearing performance while minimizing material usage for a competitive three-point bending challenge.

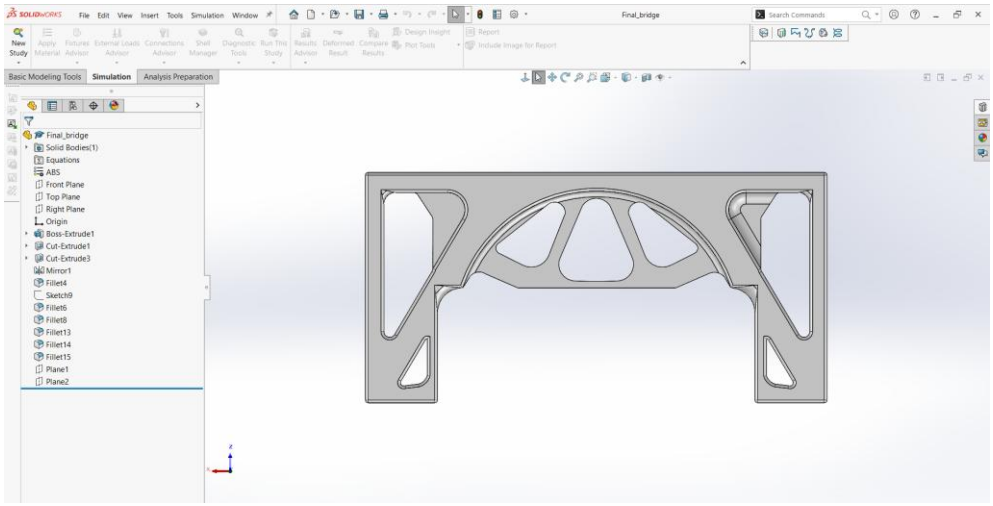
- **Design Description**
  - Developed a topology-inspired ABS bridge in SolidWorks using arched load paths, hollow cutouts, and mirrored symmetry. The upper arch was designed to redirect vertical loading into compression paths toward the side supports, reducing pure bending demand at the center span.
  - Material was removed from low-stress regions while retaining continuous ribs along the expected load paths. Fillets were added around cutouts and internal corners to reduce stress concentration and improve FDM printability. Static displacement simulation was used to evaluate deformation behavior under applied loading.

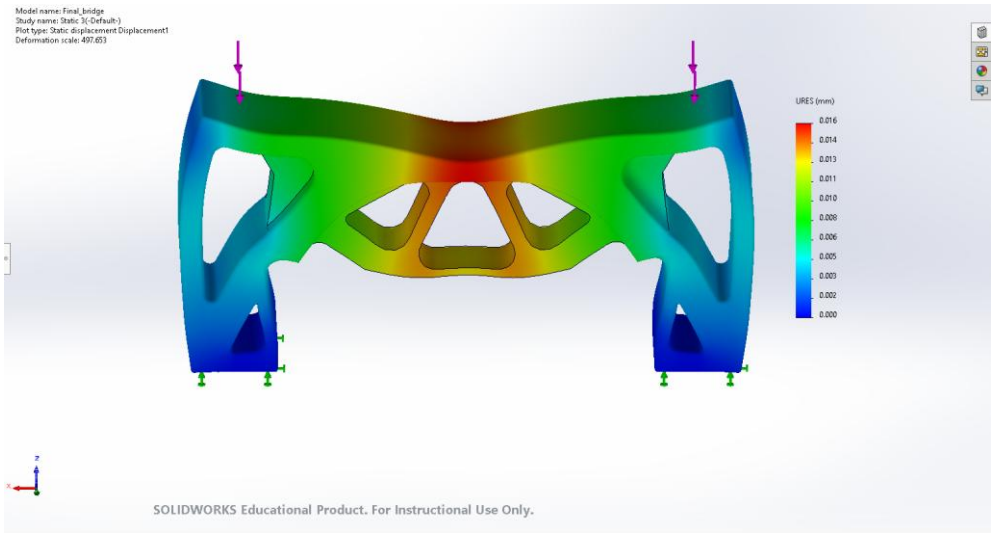
- **Key Design Features**
  - Arched upper span for compression-driven load transfer
  - Hollow topology-inspired cutouts for mass reduction
  - Mirrored symmetric geometry for balanced loading
  - Filleted cutout edges to reduce stress concentration
  - Continuous ribs retained along structural load paths
  - ABS material selected for FDM printing
  - Static displacement analysis in SolidWorks

- **Engineering Considerations**
  - Designed for quasi-static three-point bending
  - Used topology-inspired geometry to reduce unnecessary mass
  - Maintained structural continuity through ribs and side supports
  - Added fillets to reduce stress concentration near cutouts
  - Balanced load transfer using symmetric geometry
  - Considered FDM printability and avoided overly thin unsupported features









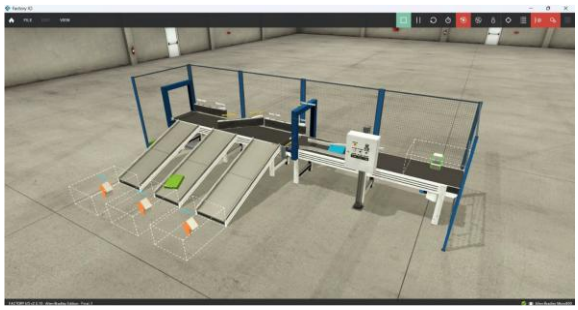
- **Result**
    - Created a lightweight bridge design with a final mass of approximately 26.35 g and volume of approximately 25,738 mm^3. The design placed 6th in the SME NCSU 3D Printing Design Challenge 2026, demonstrating strong load-to-weight performance through topology-driven structural design.

- ## Automated 3-Color Sorting Station with PLC, IIoT, and AI Perception

- **Design Type:** Smart manufacturing, PLC automation, IIoT, cloud dashboard, AI inspection
  Tools Used: Factory I/O, Micro800 PLC, CCW, Modbus TCP, MQTT, Python, PostgreSQL, Streamlit, Roboflow, YOLOv8, OpenCV

- **Objective**
  - Develop a connected smart manufacturing station that sorts blue, green, and grey boxes using PLC control, logs production data to the cloud, and uses AI perception for secondary verification.
- **Design Description**
  - Built a Factory I/O three-color sorting line controlled by an Allen-Bradley Micro800 PLC. The ladder logic included start/stop latching, conveyor control, color-based sorting, timer-driven diverter actuation, and CTU counters for each box color. PLC data was exposed through Modbus TCP, transmitted through a Python MQTT gateway, stored in PostgreSQL on Render, and visualized through a Streamlit dashboard. A YOLOv8 model trained with Roboflow-labeled Factory I/O images was added to verify box detection and sorting behavior in real time.
- **Key Design Features**
  - PLC-controlled conveyors and three diverter lanes
  - Color-based sorting logic using sensor integer values
  - CTU counters for blue, green, grey, and total count tracking
  - Modbus TCP register mapping for external data access
  - MQTT-based remote monitoring and control
  - PostgreSQL cloud database for production history
  - Streamlit dashboard for live throughput visualization
  - YOLOv8 + OpenCV perception layer for AI-based verification

- **Engineering Considerations**
  - Designed PLC logic to maintain deterministic real-time control of conveyors and sorters.
  - Used Modbus TCP to expose PLC counters, system status, and actuator states to external systems.
  - Built an MQTT publish/subscribe layer to support remote monitoring and control.
  - Stored production data in PostgreSQL for historical tracking and traceability.
  - Designed the Streamlit dashboard to visualize live throughput and box distribution.
  - Added YOLO-based perception as a secondary validation layer to cross-check PLC sorting behavior.
  - Considered scalability for additional sorting lanes, more cameras, predictive maintenance, and expanded analytics.
  - Addressed practical integration issues such as Modbus polling delay, MQTT transmission consistency, and data synchronization.



Factory I/O Scene



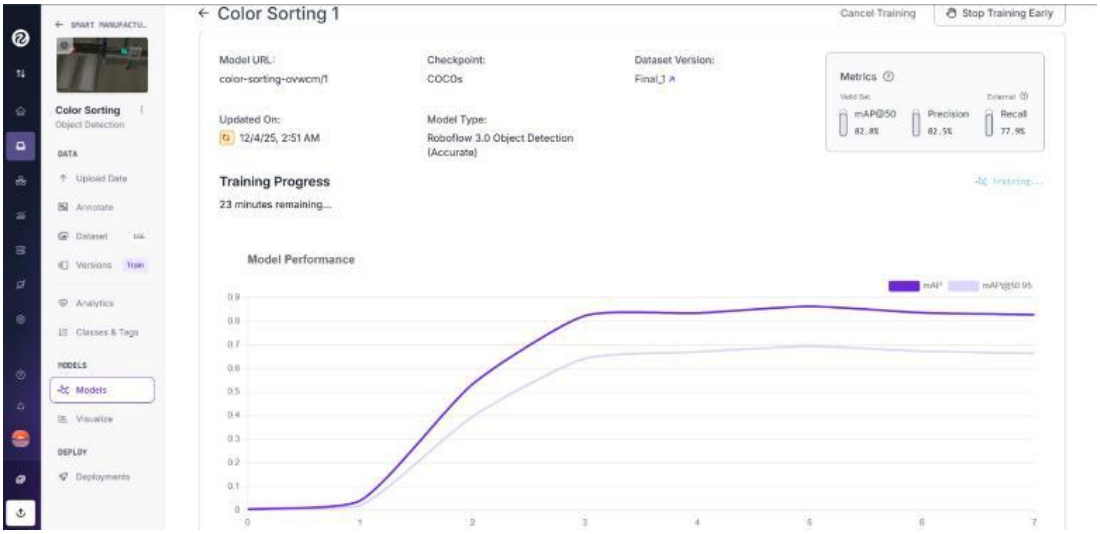
Factory I/O Drivers

BOOL_IN_1	BOOL	000001	000001
BOOL_IN_2	BOOL	000002	000002
BOOL_IN_3	BOOL	000003	000003
INT_IN_0	INT	000004	000004 - 000019
BOOL_OUT_0	BOOL	000020	000020
BOOL_OUT_7	BOOL	000027	000027
INT_OUT_0	DINT	000028	000028 - 000059
INT_OUT_1	DINT	000060	000060 - 000091
INT_OUT_2	DINT	000092	000092 - 000123
BOOL_OUT_1	BOOL	000021	000021
BOOL_OUT_2	BOOL	000022	000022
BOOL_OUT_3	BOOL	000023	000023
BOOL_OUT_4	BOOL	000024	000024
BOOL_OUT_5	BOOL	000025	000025
BOOL_OUT_6	BOOL	000026	000026

Figure 5: Modbus Mapping



Figure 9: Roboflow working using code and Factory I/O

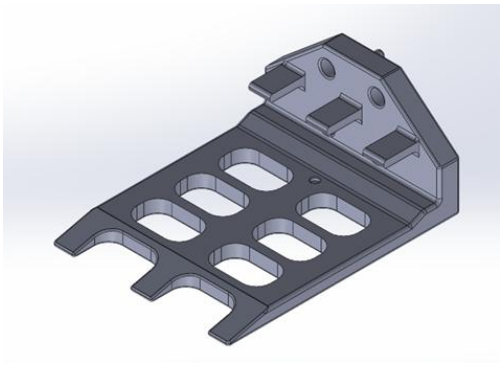


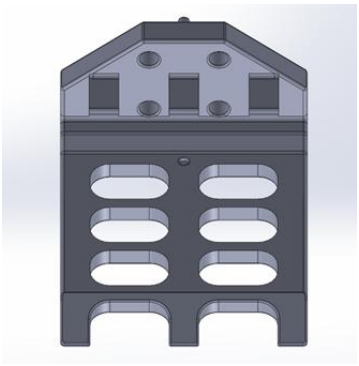
Training performance of the YOLO model on Roboflow, showing mAP (Mean Average Precision), precision, and recall trends stabilizing after several epochs

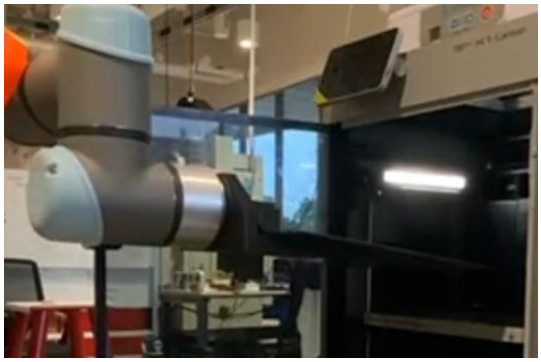
- **Result**
    - Created a complete Industry 4.0 sorting workflow integrating PLC control, IIoT communication, cloud analytics, and AI perception. The system demonstrated stable real-time sorting and live dashboard monitoring, while the YOLOv8 model achieved **82.8% mAP@50**, **82.5% precision**, and **77.9% recall**.

- ## 3D-Printed UR5e End-Effector Tool for Print Bed Removal

- **Design Type:** Robotic end-effector, FDM tooling, automation fixture
  Software Used: SolidWorks, Bambu Slicer
- **Objective**
  - Design a custom 3D-printed end-effector tool for a UR5e robot manipulator to lift and remove a 3D printer print bed during an automated handling workflow.
- **Design Description**
  - Designed a single-piece FDM-printable robotic attachment with a fork-style front profile for print bed engagement and a rear mounting plate for robot-side connection. The body includes elongated lightweight cutouts to reduce mass while maintaining structural ribs for stiffness.
  - The rear mounting face includes bolt holes and alignment geometry for attachment to the UR5e tool setup. Rounded slots and filleted internal edges were used to reduce stress concentration and improve printability. The front geometry was shaped to engage the print bed while avoiding interference with the printer frame.
- **Key Design Features**
  - Fork-style front geometry for print bed lifting/removal
  - Rear mounting interface with bolt holes
  - Lightweight cutouts to reduce robot payload demand
  - Rounded slots and fillets to reduce stress concentration
  - Single-piece FDM-printable structure
  - Wide contact area for stable bed handling
  - Robot-compatible geometry for repeatable automation
- **Engineering Considerations**
  - Reduced unnecessary material to lower robot payload demand
  - Maintained ribs between cutouts to preserve stiffness
  - Designed mounting holes for robot-side attachment
  - Considered clearance between the tool, print bed, and printer frame
  - Used FDM printing for fast prototyping
  - Designed for repeatable robotic handling instead of manual removal



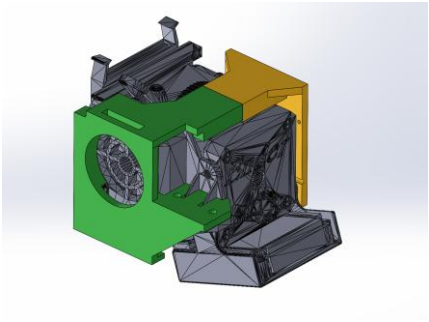
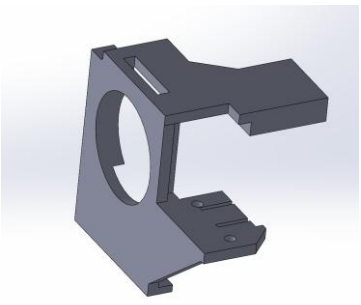
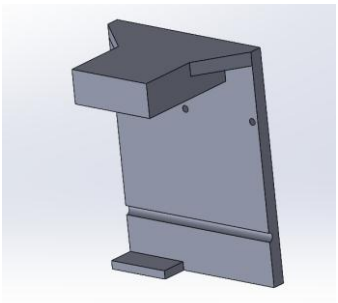




- **Result**
  - Fabricated and tested a functional 3D-printed end-effector that enabled UR5e-compatible print bed removal and demonstrated custom FDM tooling for lab automation.

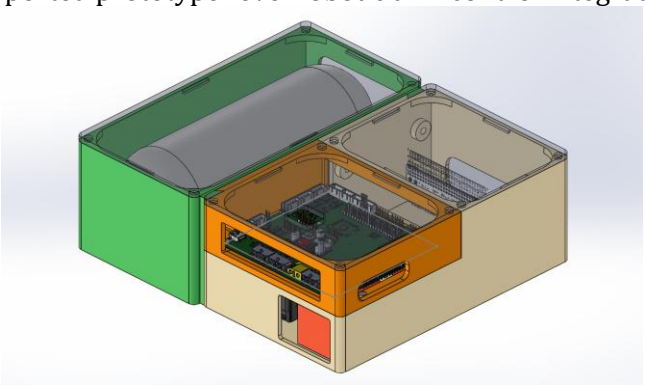
- ## Ender 3D Printer Nozzle Enclosure for UR5e Robotic Tool Integration

- **Design Type:** Robotic tool enclosure, FDM-printed mechanical housing, nozzle/hotend integration
  Tools Used: SolidWorks, Bambu Slicer
- **Objective**:
  Design a custom 3D-printable enclosure to package and support a 3D printer nozzle assembly so it could be integrated as a tool attachment for a UR5e robotic system.
- **Design Description**
  - Developed a multi-component mechanical enclosure around an existing 3D printer nozzle/hotend assembly. The design included a main support housing, nozzle clearance opening, rear mounting plate, and front guide/enclosure section. The enclosure was split into multiple printable components to simplify fabrication, assembly, and maintenance.
  - The geometry-maintained access to the nozzle region while supporting and protecting the surrounding hardware. The design also provided mounting surfaces for integration with the robot-side tool structure.
- **Key Design Features**
  - Multi-part FDM-printable enclosure
  - Nozzle clearance opening to prevent mechanical interference
  - Rear mounting plate for tool-side integration
  - Open-frame geometry for access and reduced material use
  - Component packaging around an existing nozzle assembly
  - Assembly-friendly structure for installation and maintenance
  - Mounting features for robotic tool integration
- **Engineering Considerations**
  - Maintained clearance around the nozzle/hotend area
  - Designed around an existing printer nozzle assembly
  - Split the enclosure into printable components
  - Considered wiring access and serviceability
  - Added mounting features for future robotic tool attachment
  - Used open geometry to reduce material while preserving access
- **Result**:
  - Fabricated a functional FDM-printed enclosure that packaged the 3D printer nozzle assembly for robotic tool integration while maintaining clearance, accessibility, and manufacturability.

- ## 3D-Printed Control Box Assembly for Exoskeleton Robotic Arm

- **Design Type:** Electronics enclosure, robotic control system packaging, FDM prototype
  Tools Used: SolidWorks, Bambu Slicer
- **Objective**:
  Design a compact 3D-printed control box assembly to house and organize electrical components used for controlling the shoulder joint movement of an exoskeleton robotic arm.
- **Design Description**
  - Designed a multi-component enclosure with internal mounting regions, rounded internal transitions, and cable/tube exit features. The internal layout was developed to organize electrical components while allowing access for assembly, wiring, inspection, and future modification.
  - The control box was constrained to a compact **260 × 226 × 90 mm** boundary box and designed for FDM fabrication. The enclosure geometry balanced space efficiency, accessibility, and printability.
- **Key Design Features**
  - Internal mounting bosses for electrical components
  - Cable/tube outlet features for external connections
  - Rounded internal corners for printability and stress reduction
  - Compact 260 × 226 × 90 mm enclosure size
  - FDM-printable housing geometry
  - Assembly-accessible internal layout
  - Multi-component control hardware package
- **Engineering Considerations**
  - Organized internal space for electrical components
  - Included cable/tube exit features for external routing
  - Maintained access for assembly and maintenance
  - Used rounded corners to improve print quality and reduce stress concentration
  - Designed within a compact boundary box
  - Supported prototype-level robotic arm control integration

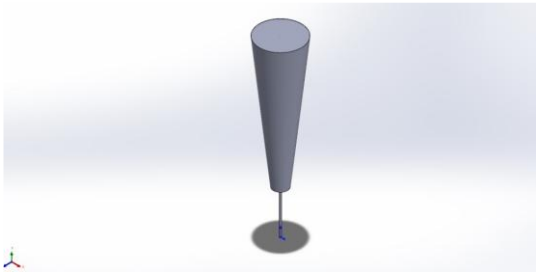


- **Result**:
  - Designed and fabricated a functional 3D-printed control box assembly for exoskeleton control hardware integration, supporting prototype-level shoulder-joint control system development.

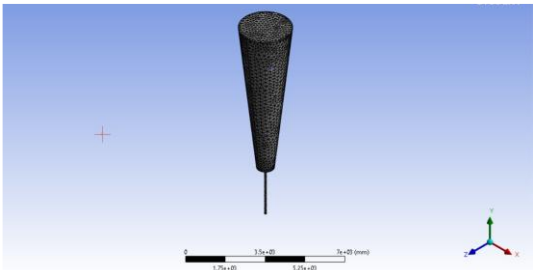
- ## Structural and Numerical Analysis of Bladeless Wind Turbine (UG Capstone)

- **Design Type:** Simulation-driven mechanical design, CFD analysis, structural analysis
  Tools Used: SolidWorks, ANSYS, MS Excel

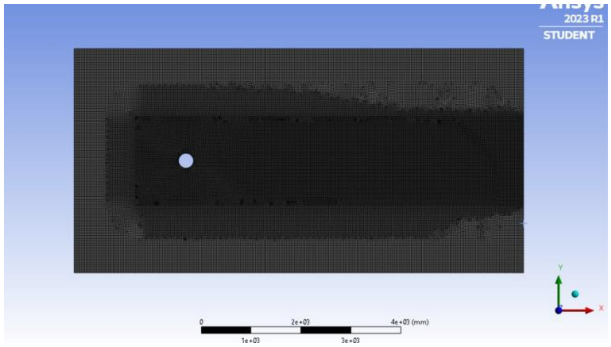
- **Objective:**
  - Design and analyze a vortex bladeless wind turbine mast to study vortex-induced oscillation behavior caused by aerodynamic lift and drag forces.
- **Design Description:**
  - Designed and modeled a vortex bladeless wind turbine structure in SolidWorks and analyzed it using ANSYS. CFD simulations were used to study velocity flow, vortex formation, and lift/drag behavior. Structural analysis was used to evaluate mast deformation under aerodynamic loading and confirm whether the system could oscillate without an unsafe structural response.
  - The turbine operates based on vortex shedding, where alternating aerodynamic forces act on the mast and induce oscillatory motion. CFD results were used to visualize wake formation and extract aerodynamic behavior over time.
- **Key Analysis Features:**
  - SolidWorks model of VBWT mast
  - ANSYS mesh generation for structural and CFD analysis
  - CFD domain for velocity flow visualization
  - Vortex formation analysis behind the mast
  - Lift and drag coefficient extraction
  - Structural deformation evaluation under wind loading
- **Engineering Considerations:**
  - Used CFD to verify vortex formation behind the mast
  - Refined mesh around the mast to better capture wake behavior
  - Monitored lift and drag coefficient variation over time
  - Evaluated mast deformation under aerodynamic loading
  - Connected aerodynamic force behavior to structural response
  - Used simulation results to support the feasibility of vortex-induced oscillation
- **Results:**
  - Simulation results showed Cd ≈ 8.4, Cl ≈ 8.18, and approximately 0.57 m deformation at 15.6 m/s inlet velocity.
  - The project was presented and published at the E2A Emerging Electronics and Automation Conference 2023.

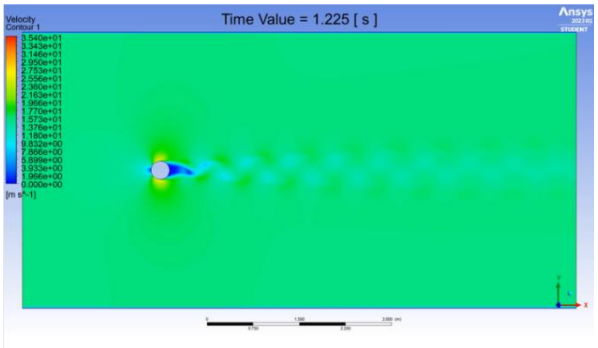


Structure of VBWT

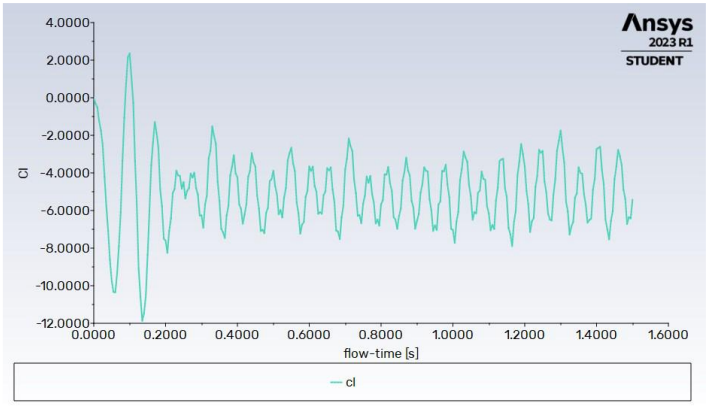


Mesh of VBWT in ANSYS





CFD Mesh of VBWT in 2D with velocity flow Vortex formation by lift force



Cd and Cl graph from ANSYS

- ## Comparative Thermal Analysis of Heat Sink Fin Geometries using ANSYS

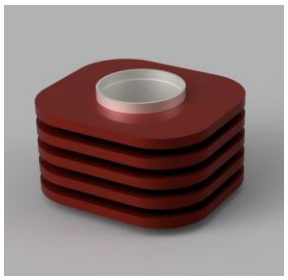
- **Design Type:** Thermal simulation, heat sink design, material comparison
  Tools Used: SolidWorks, ANSYS, MS Excel

- **Objective:**
  - Analyze multiple heat sink fin geometries to determine the most effective design for heat dissipation under identical thermal loading conditions.

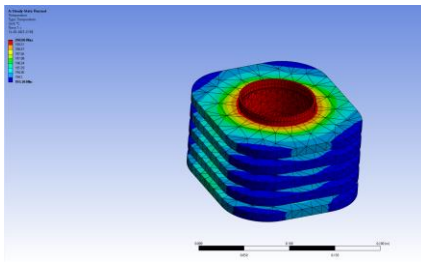
- **Design Description:**
  - Designed multiple fin geometries in SolidWorks and evaluated their thermal performance using ANSYS steady-state thermal analysis. The study compared straight fins with curved edges, triangular fins, pin-style fins, and rounded-edge straight fins across stainless steel, grey cast iron, and copper.
  - Temperature contours and Excel-based temperature-fall graphs were used to compare thermal behavior, fin effectiveness, and fin efficiency. The project focused on understanding how geometry and material selection influence heat dissipation performance.

- **Key Design Features:**
  - Multiple fin geometries modeled and compared
  - Material comparison across stainless steel, grey cast iron, and copper
  - ANSYS thermal contour post-processing
  - Temperature-fall graphs generated in Excel
  - Effectiveness and efficiency comparison under identical conditions
  - Geometry-based evaluation of heat dissipation performance

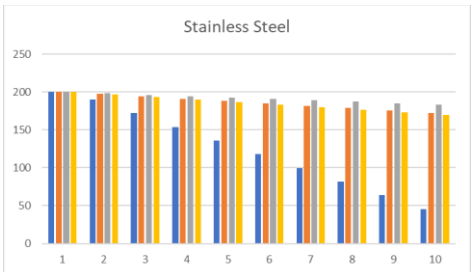
- **Engineering Consideration:**
  - Compared the effect of fin spacing, edge curvature, and surface area
  - Maintained consistent thermal loading for fair comparison
  - Studied how material thermal properties affected temperature drop
  - Used ANSYS contours to identify heat concentration regions
  - Used Excel graphs to compare temperature fall across materials
  - Evaluated both effectiveness and efficiency, not just maximum temperature



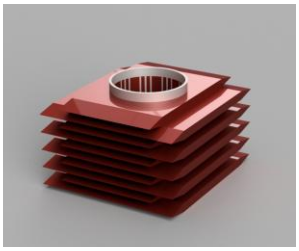
Straight Fin with Curved Edge



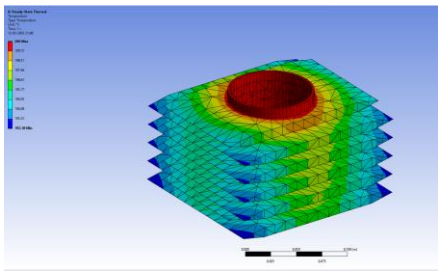
Thermal Analysis in ANSYS



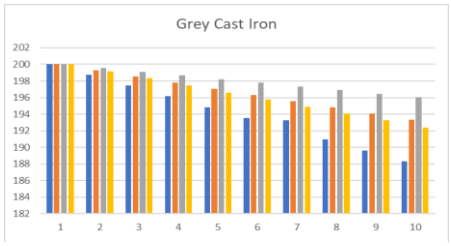
Temperature fall graph - SS



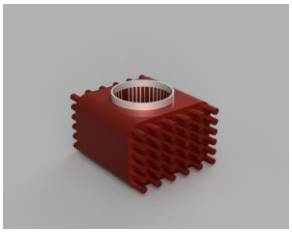
3D Model of Triangular Fin



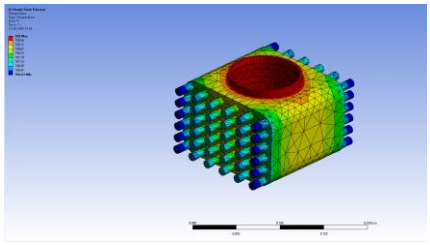
Thermal Analysis in ANSYS



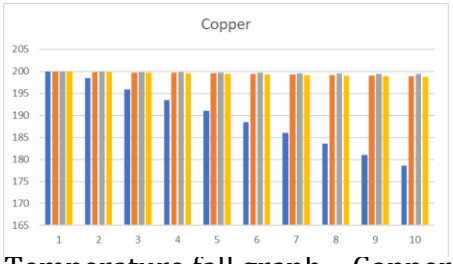
Temp fall graph – Grey Cast Iron



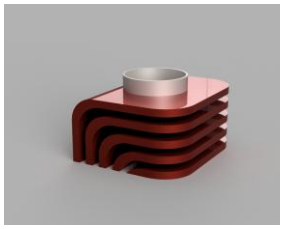
3D Model of Triangular Fin



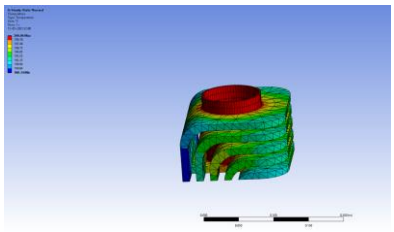
Thermal Analysis in ANSYS



Temperature fall graph – Copper



Straight Fin with Round Edge



Thermal Analysis in ANSYS

- **Results:**
  - Pin fins were approximately **57% more effective than straight fins**, while straight fins with rounded edges showed the highest efficiency due to improved airflow behavior and reduced sharp-edge flow disruption.